

FAIR2: A framework for integrating community knowledge in data analytics and addressing discrimination bias

Centering the Public Interest in Integrated Administrative Data Analytics

Module 1 of 5

Public Interest Technology University Network Year 6 Challenge
January 2026



Module 1

FAIR2: A framework for integrating community knowledge in data analytics and addressing discrimination bias

Module 2

Demographic Data in Integrated Data Systems: Seeking Experiential Knowledge to Inform Race and Gender Data



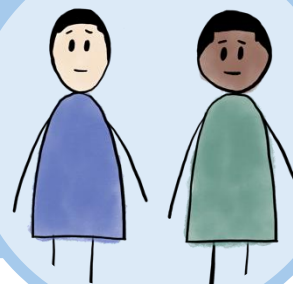
Module 3

An illustration of Label Bias in Homelessness Data Analytics



Module 4

An illustration of Collider Bias in housing research



Module 5

Recommendations to address constraints and bias in Integrated Data Analytics



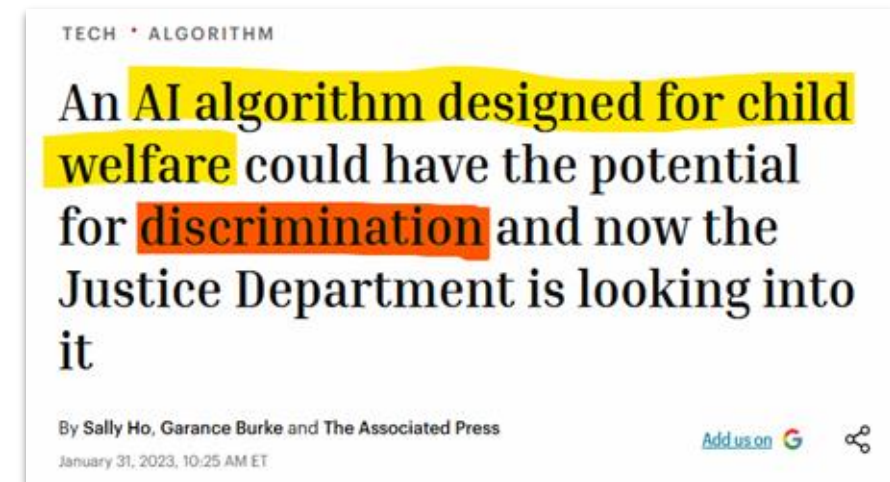
Public interest in an Era of Big Data Technologies

Decision making in health and human services is **increasingly aided by data technologies**.

How do we ensure that data analytics and metadata account for the experiences of those represented and avoid perpetuating bias in society?



(Tajo et al., 2024)



(Goldstein, 2023)



(Gawronski, 2019)

What is FAIR2?



Ethical standards of metadata FAIRification: **F**indable, **A**ccessible, **I**nteroperable and **R**eusable principles (Wilkinson et al., 2016)

When using social **data for the public interest**, *add* FAIR2 principles (Richter et al., 2023):

- **F**rame meta(data) with contextual, experiential knowledge
- **A**rticulate assumptions into system (causal) graphs
- **I**dentify and address bias in data analytics
- **R**eport findings to community stakeholders and seek feedback.

FAIR2 is a framework for Public Interest Technology (PIT) in Data Analytics

PIT: study and application of tech expertise to advance the public interest - generate public benefits - promote the public good. <https://cwru-dsci.org/>

Why FAIR2? Predicting High-Risk Patients to Increase Health Care Support

(Obermeyer et al., 2019)

- **Analytics:** A major health system used an algorithm to flag patients with the most complex needs to enroll them in a specialized care management program. Analysts used **Total Health Care Costs** as a proxy for **Medical Need**. **They assumed** that the more a person spends on healthcare, the worse their health condition
- **Result:** at **the same data-derived risk score**, African American patients had more severe diabetes, kidney failure, etc., than white patients
- **Data** most likely agreed with **Standard FAIR principles**: it was *Findable, Accessible, Interoperable*, and *Reusable*. **So, how did bias arise?** Structural inequities (lower access to care, bias in treatment), meant the system historically **spent less** on Black patients
- **Under the FAIR2 Framework**, experiential knowledge of patients would have clarified that cost data are not a good reflection of health status for patients historically marginalized

FAIR2 draws from work by scholars and activists calling for better data science now and then



Fannie B. Williams



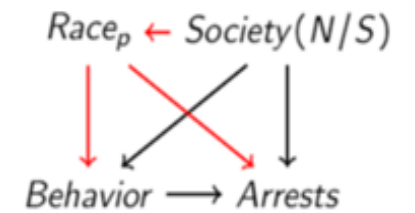
W. E. B. Du Bois



Kelly Miller



Ida B. Wells



Muhammad, K. G. (2019), Pearl, J., & Mackenzie, D. (2018). Kleinberg et al, (2018), Howe et al., 2022.

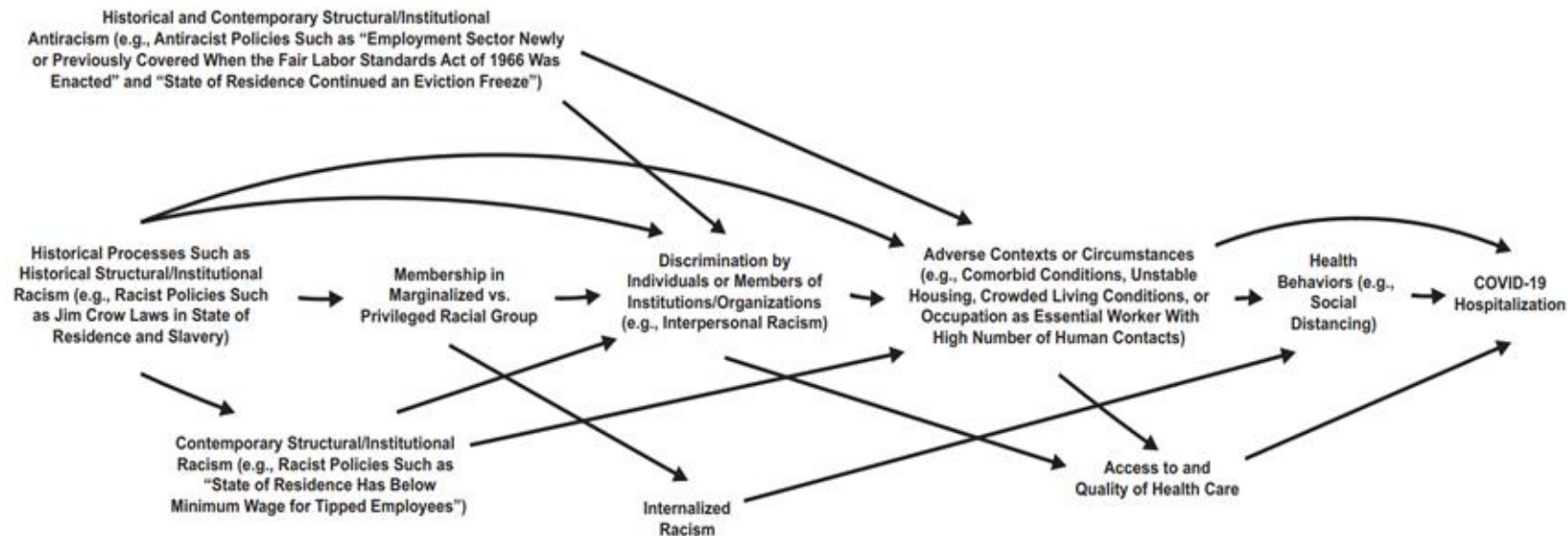
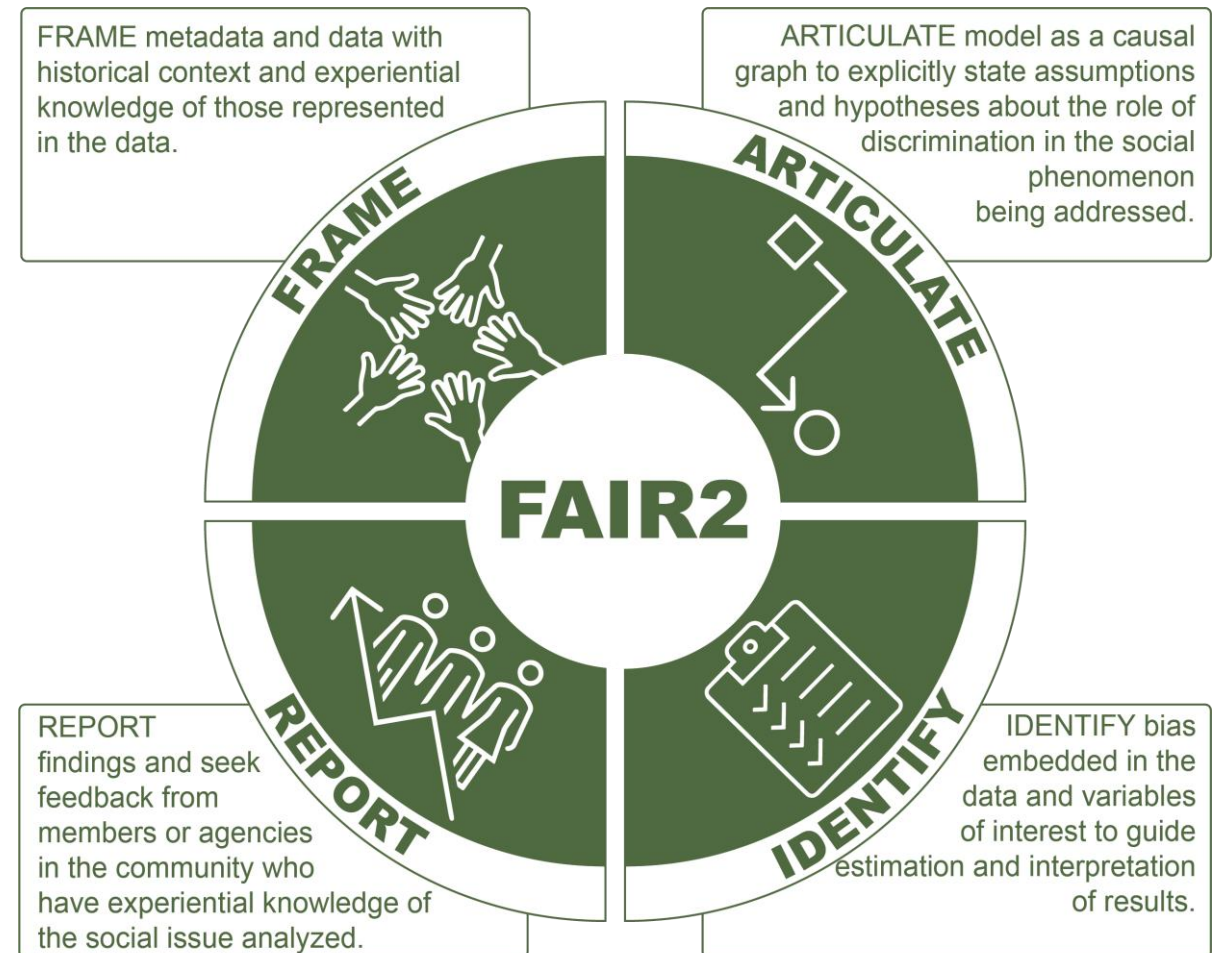


Figure 2. Modified version of Figure 1 that can inform the answer to a particular research question, in this case, the hypothetical impact of a specific intervention on racial disparities in coronavirus disease 2019 (COVID-19) hospitalizations.

(Howe et al., 2022)

FAIR2 recognizes that data do not speak for themselves, that observational data reflect discrimination in society, and that an explicit framework to identify and address discrimination biases will further data science's potential to advance social justice.

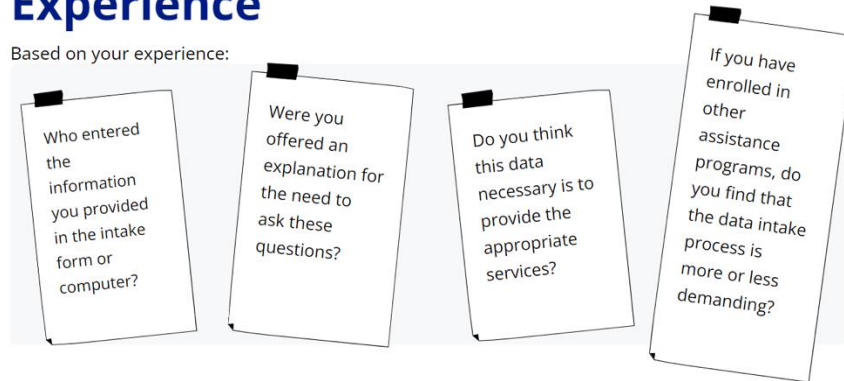


Frame metadata and data with historical context and experiential knowledge of those represented in the data: **Data Chats**



Data Intake Experience

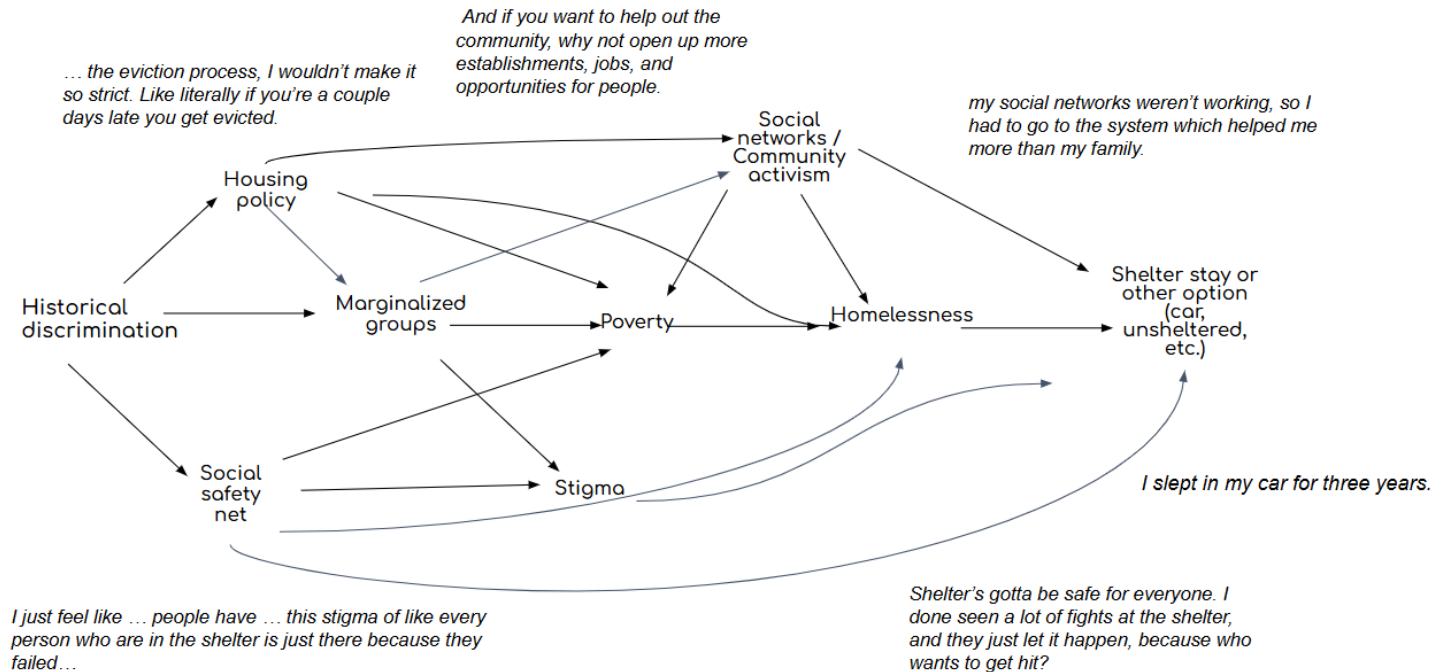
Based on your experience:



- Data Chats are guided conversations between practitioners and people whose experiences are represented in the data
- Aim is to leverage experiential knowledge to inform the inputs of the data analytic process or research activity

Articulate the general model as a causal graph

What causes young people to experience homelessness and shelter stay?



- Use insights from **Data Chats** and the literature to build a causal model
- A Causal Graph maps assumptions about how the process that generated the data we want to analyze
- Aim is to **explicitly state model assumptions (background knowledge) and hypotheses about the role of discrimination in the social phenomenon studied**

Identify biases embedded in the data and variables of interest



What is the social goal?	Prioritize housing for those at highest risk of homelessness. Data analytics aim is to predict homelessness risk.
Choice of the Outcome	Desired Y: unsheltered at 6 months. Measured Y: seen in administrative homeless service (HMIS) records at 6 months. Bias when using measured versus desired Y: if previous shelter experience was negative, folks will experience homelessness outside of the system and not appear in administrative records.
Choice of the Inputs	Desired X: experienced food insecurity in past year Measured X: enrolled in SNAP program as seen in administrative records Bias when using measured versus desired X: food insecure folks that do not meet work requirements will not be in SNAP and assumed food secure. Role of variable reflecting discrimination status (sex, gender, race, ethnicity)
Data	Selective Labels: if we only use HMIS records, we will fail to predict homelessness for vulnerable groups that feel unsafe in shelter. Race and gender are captured in diverse ways in administrative data; how do we aggregate and report along these categories?

Approach adapted from Kleinberg et al. (2018)

- Systematically scan data inputs for mismatches in desired versus measured constructs and target population
- Aim is to minimize bias and report on limitations due to bias

Report findings to and seek feedback from community stakeholders



- Feedback from various Data Chat participants after receiving report:

I'm just so glad you all are taking your time to hear our thoughts and experiences. I feel like students using this for research should not be the final destination ... We need to show this to people in charge ...

"I feel like people are eager to share their experience if people are willing to listen."



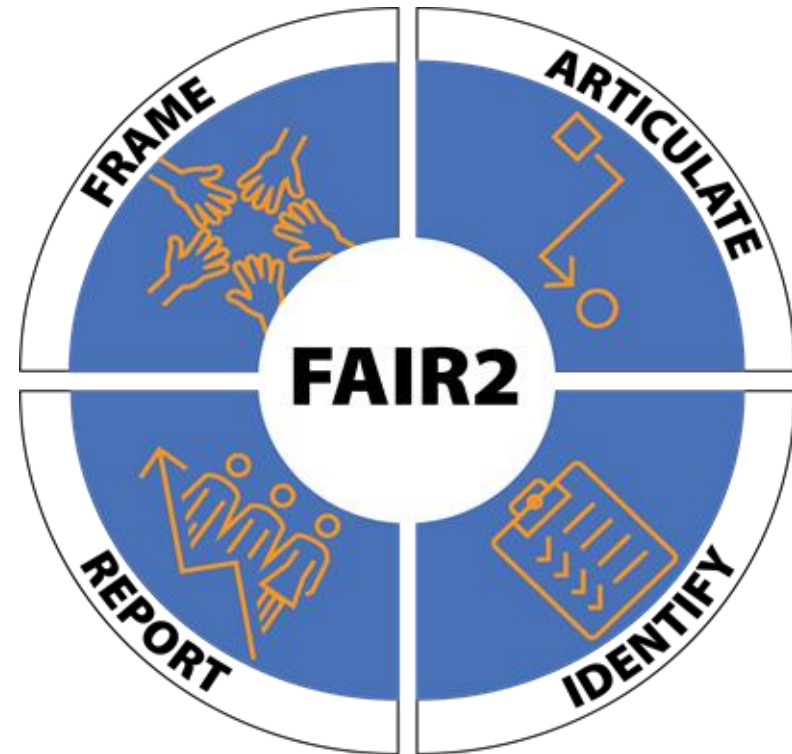
"Statistics are important, but I think we should spend more time emphasizing the *why* versus the *who*."

- Report back to Data Chat participants and stakeholders
- Aim is to democratize the data analytic process, seeking feedback from community members whose well-being is at stake and from agencies who work to address the social issue analyzed

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References

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